

AI-Based Diabetes Diagnosis System

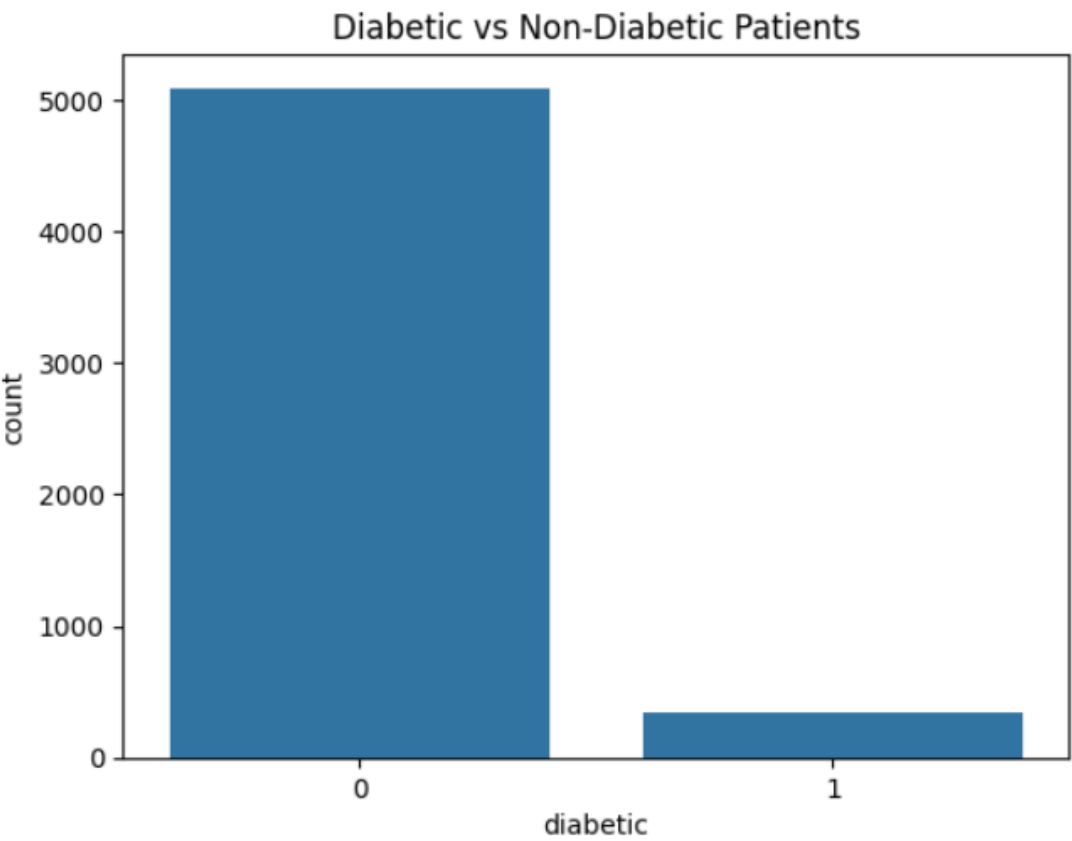
Overview

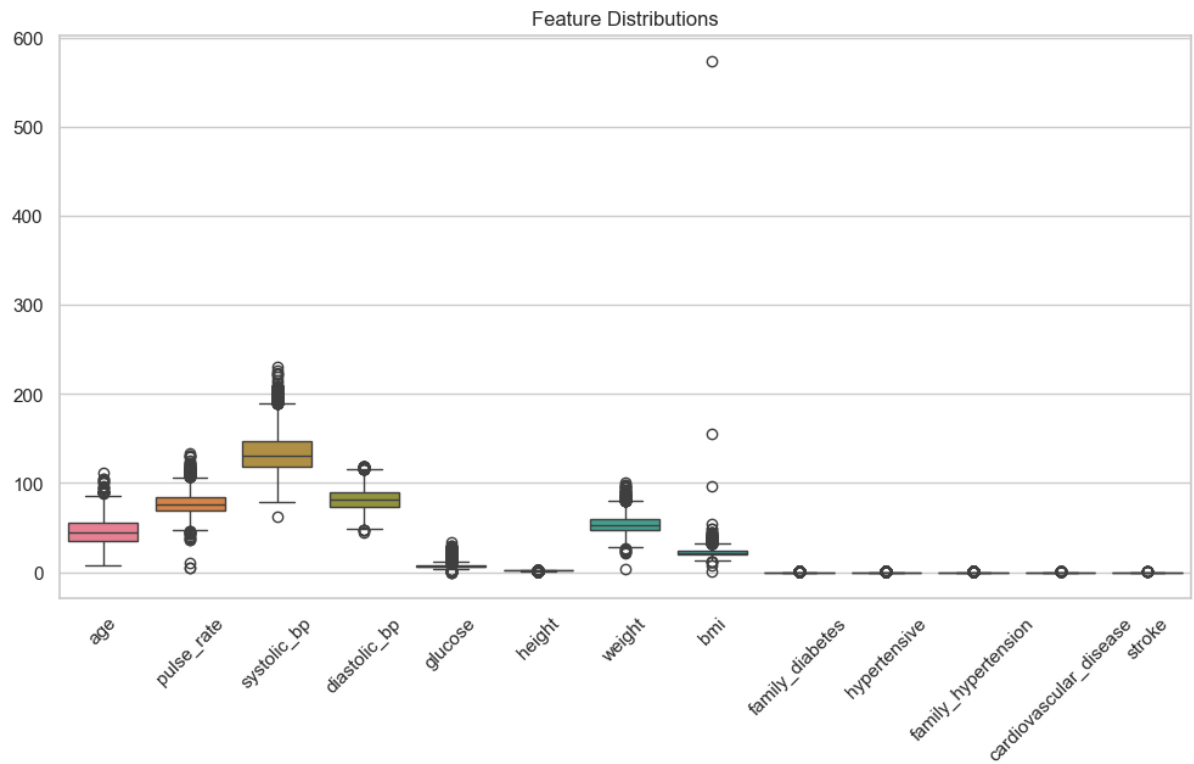
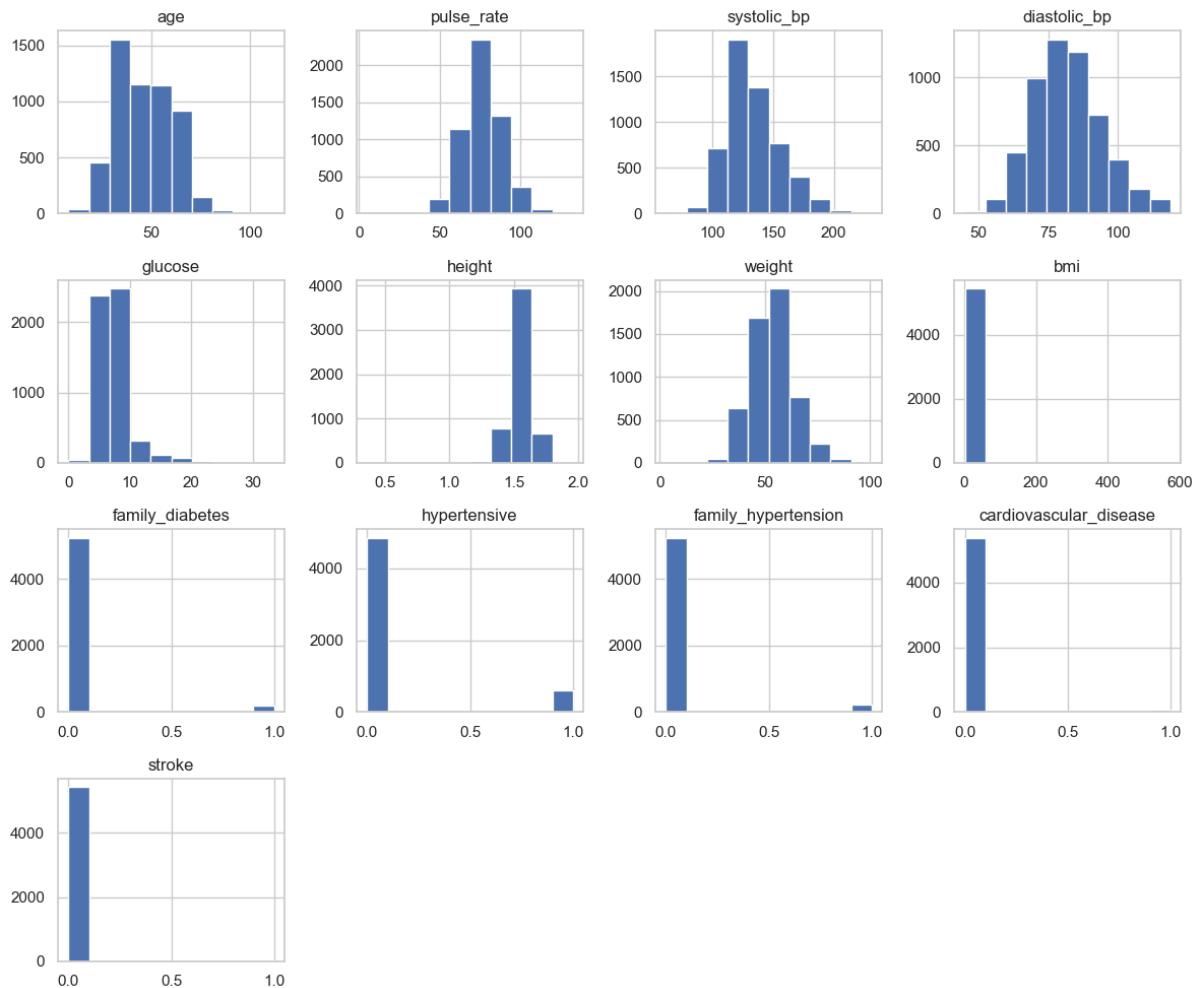
This project develops an end-to-end AI system for diabetes diagnosis using machine learning and large language models (LLMs). The solution integrates predictive modeling with conversational AI for comprehensive patient assessment.

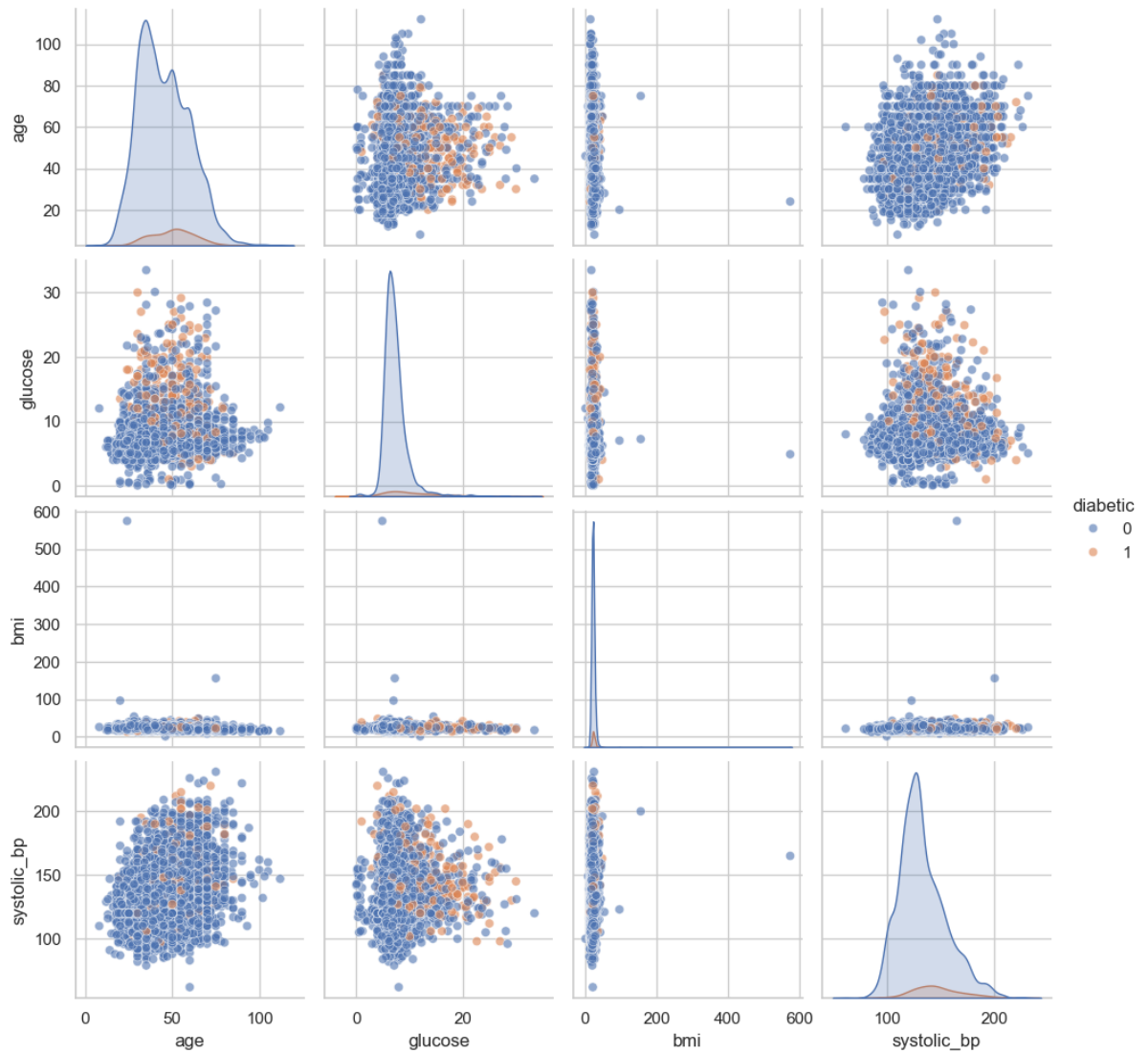
Data Processing

Initial Dataset

- **5,000 patient records** with diabetic/non-diabetic classification
- Significant class imbalance:
 - Non-diabetic (0): ~4,500 cases
 - Diabetic (1): ~500 cases

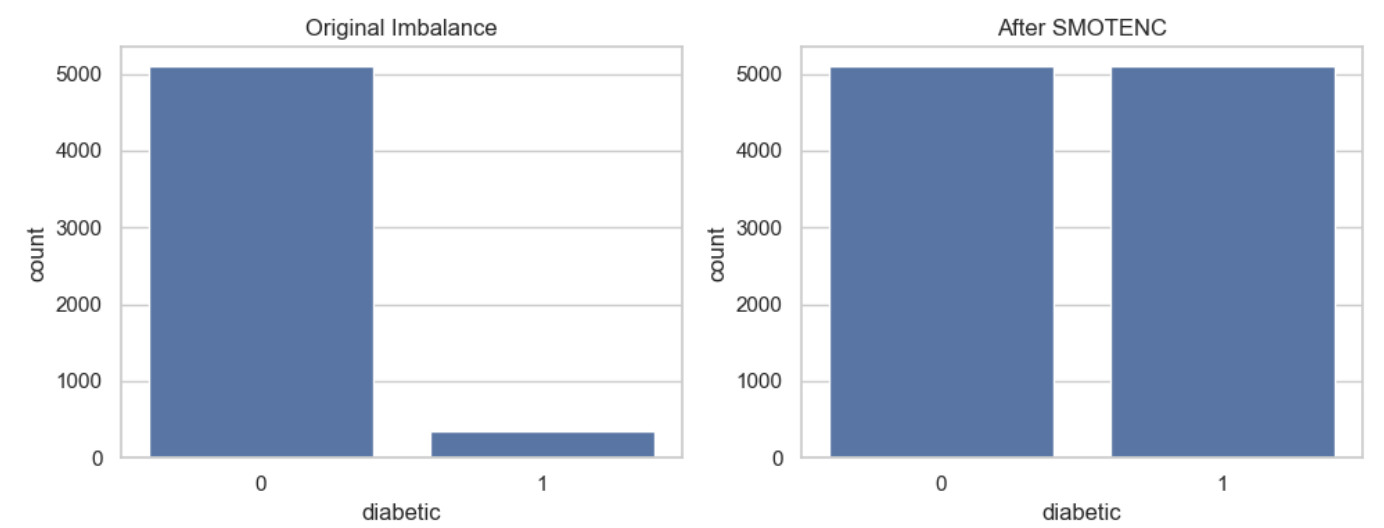






Preprocessing Steps

1. Removed invalid/zero values from critical features
2. Eliminated physiologically impossible values
3. Applied **SMOTENC** for class balancing:
 - Handles mixed data types (numerical + categorical)
 - Generates synthetic minority class samples
 - Preserves feature relationships



Class balance before (left) and after (right) SMOTENC application

Model Development

Algorithms Tested

- 1. Logistic Regression
- 2. Random Forest
- 3. XGBoost
- 4. LightGBM
- 5. Neural Network (2-layer)

Training Approach

- Hyperparameter tuning via GridSearchCV/Optuna
- 5-fold cross validation
- ROC AUC as primary metric

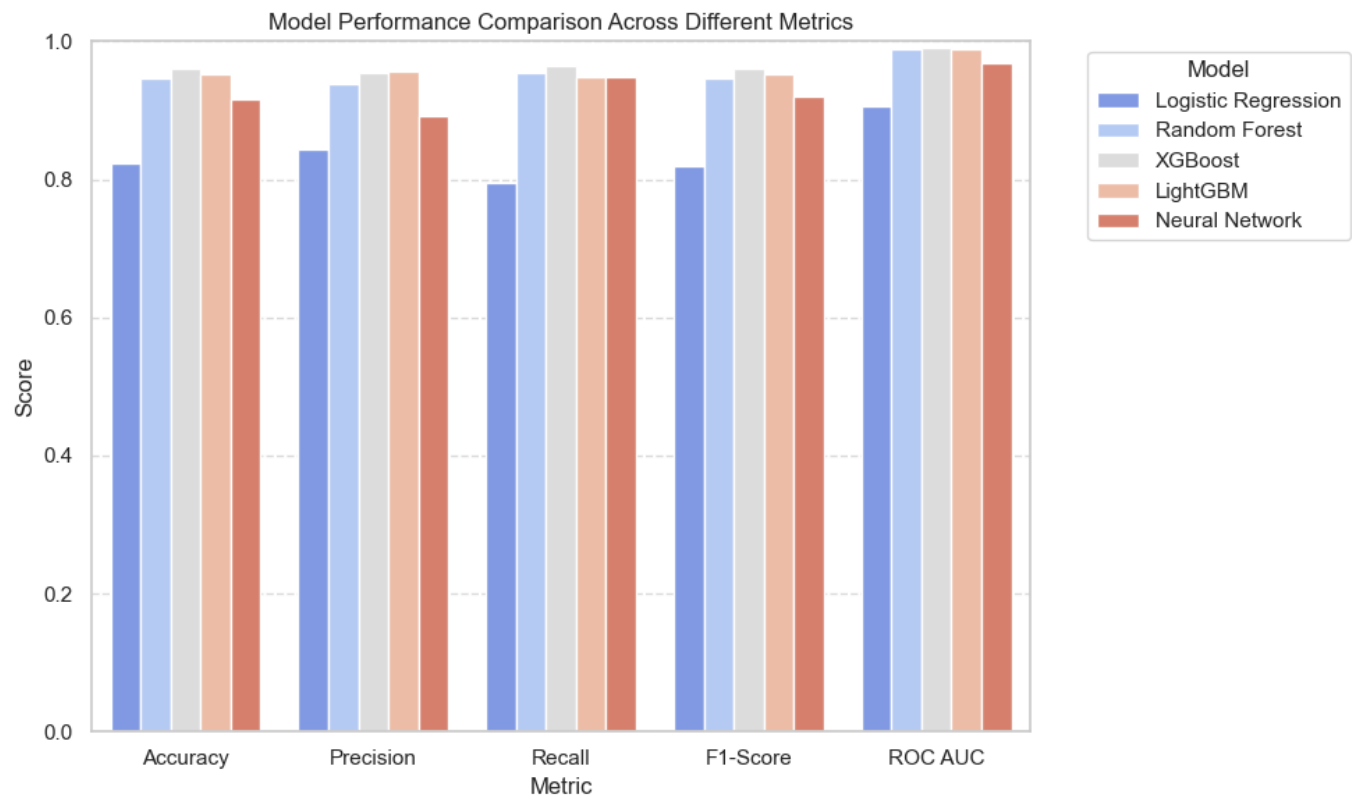
```
# Hyperparameter Tuning Example
param_grid = {
    'max_depth': [3, 5],
    'learning_rate': [0.01, 0.1]
}
model = GridSearchCV(XGBClassifier(), param_grid, cv=5, scoring='roc_auc')
model.fit(X_train, y_train)
```

Performance Results

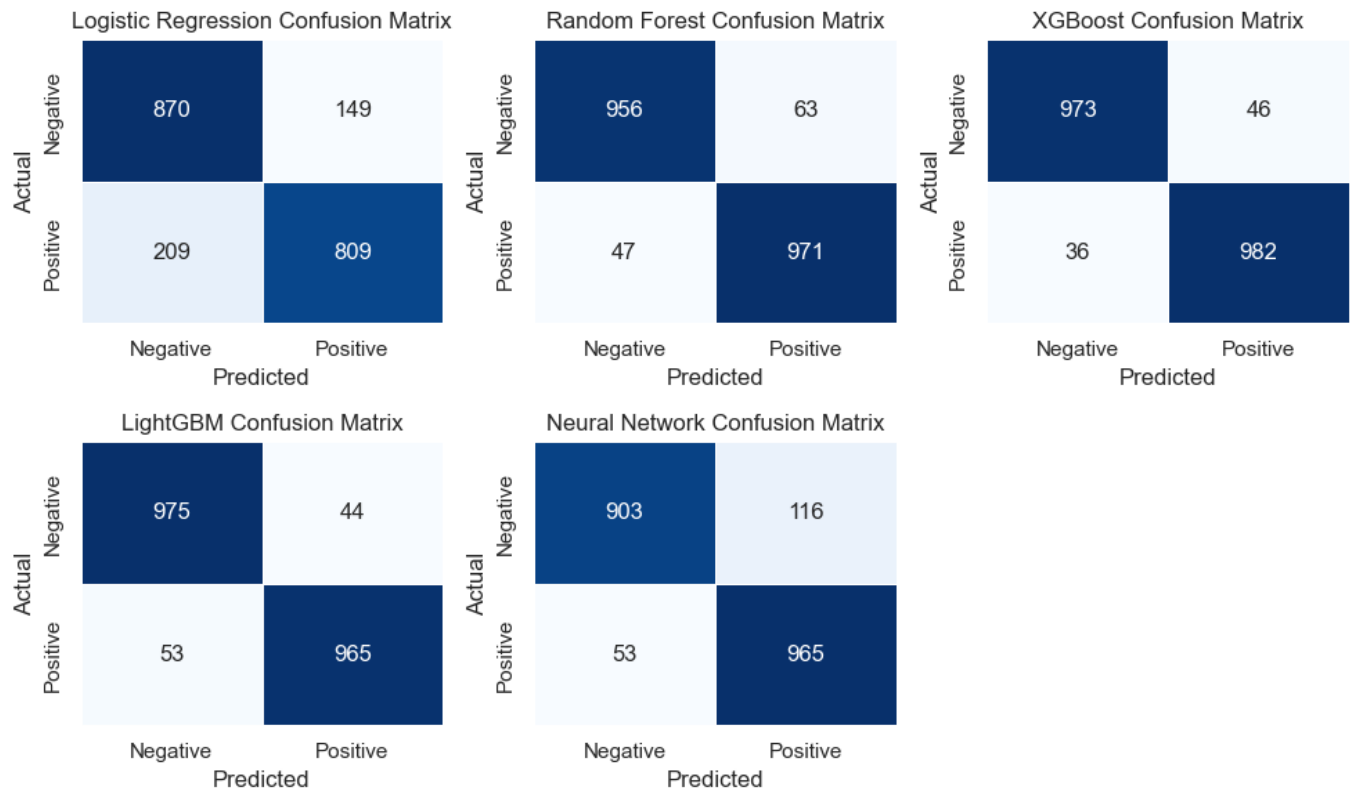
Model Comparison

Metric	XGBoost	LightGBM	Random Forest	Logistic Reg	Neural Net
Accuracy	95.97%	94.2%	93.5%	89.8%	91.3%
ROC AUC	0.982	0.974	0.968	0.932	0.947

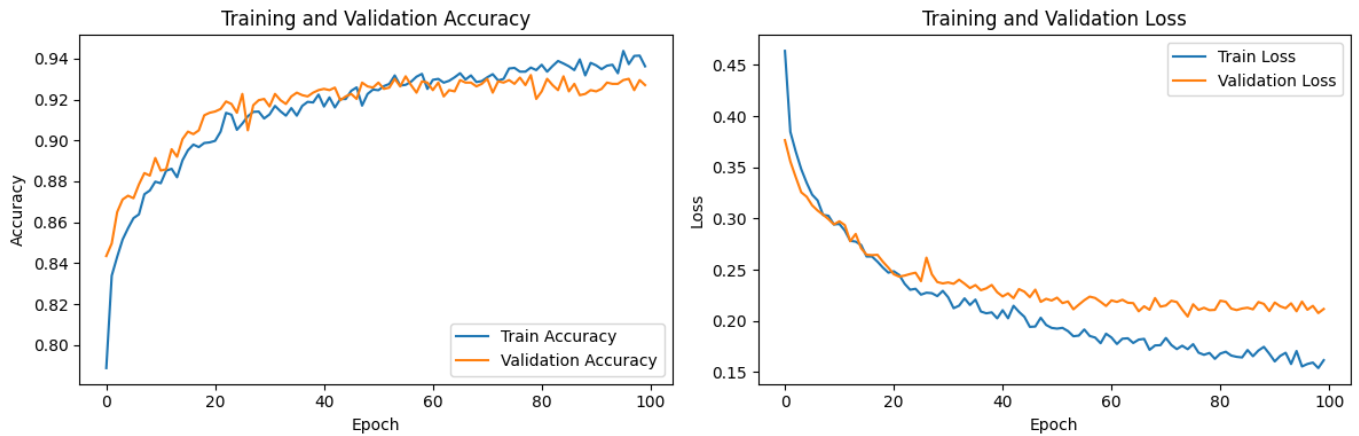
Metric	XGBoost	LightGBM	Random Forest	Logistic Reg	Neural Net
F1-Score	0.927	0.914	0.901	0.856	0.879



XGBoost achieved highest accuracy (95.97%)



Neural Network Training



Optimal parameters: 64 nodes, 0.2 dropout, 0.005 learning rate, 128 batch size

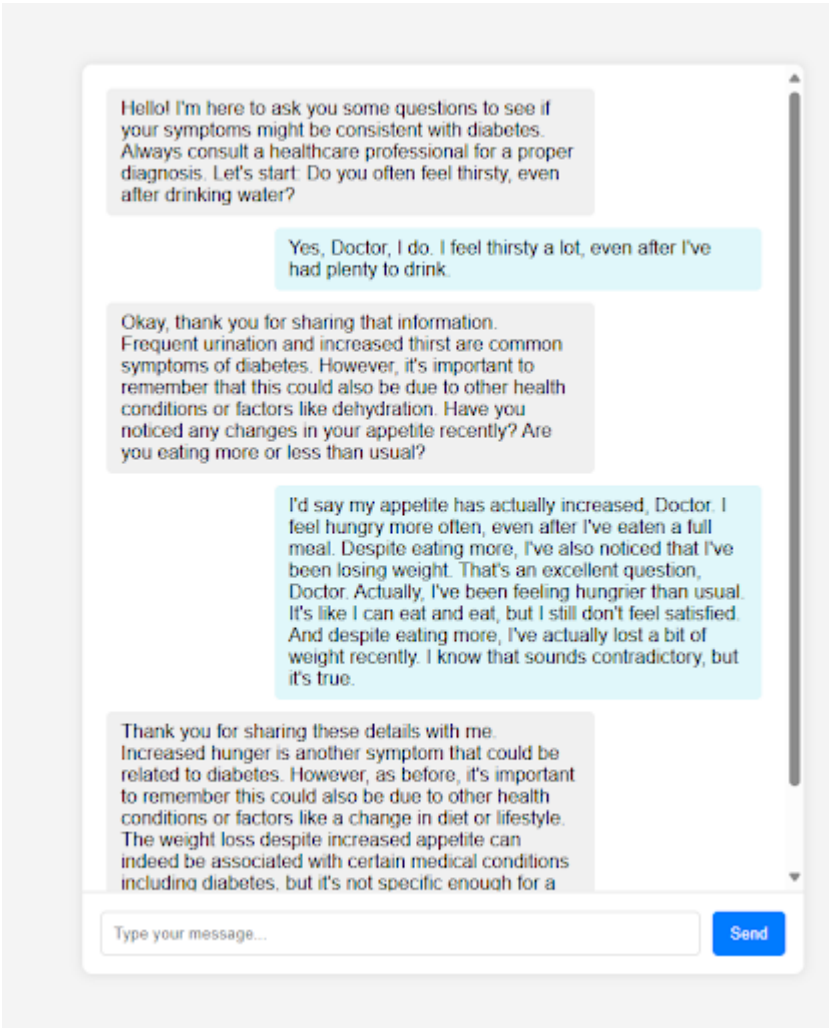
AI Diagnostic Agent

Dual-Agent Architecture

1. **Conversational Agent (LLM-Based):**

- Uses LLaMA 3.2 with medical prompt engineering
- Conducts symptom-based interview
- Triggers prediction API when confidence >25%

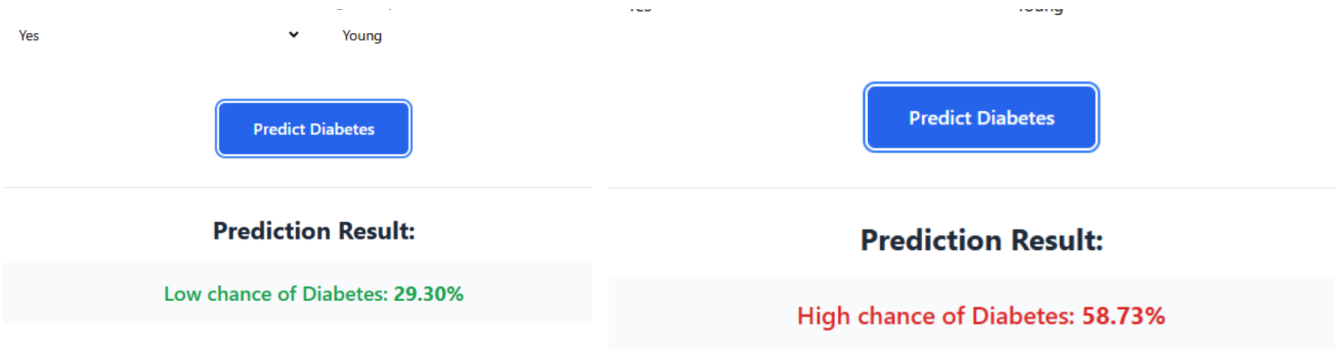
"Hello! I'm here to ask questions about potential diabetes symptoms.
Do you experience excessive thirst even after drinking water?"



2. Prediction Agent:

- Processes 15+ clinical features:
 - Demographics (age, gender)
 - Vital signs (BP, glucose, BMI)
 - Medical history (hypertension, stroke)
 - Family history

Prediction Interface



Clinical input form with real-time risk assessment

Sample Output:

"Low chance: 29.30% | High chance: 58.73%"

Explainable AI (XAI)

Interpretation Techniques

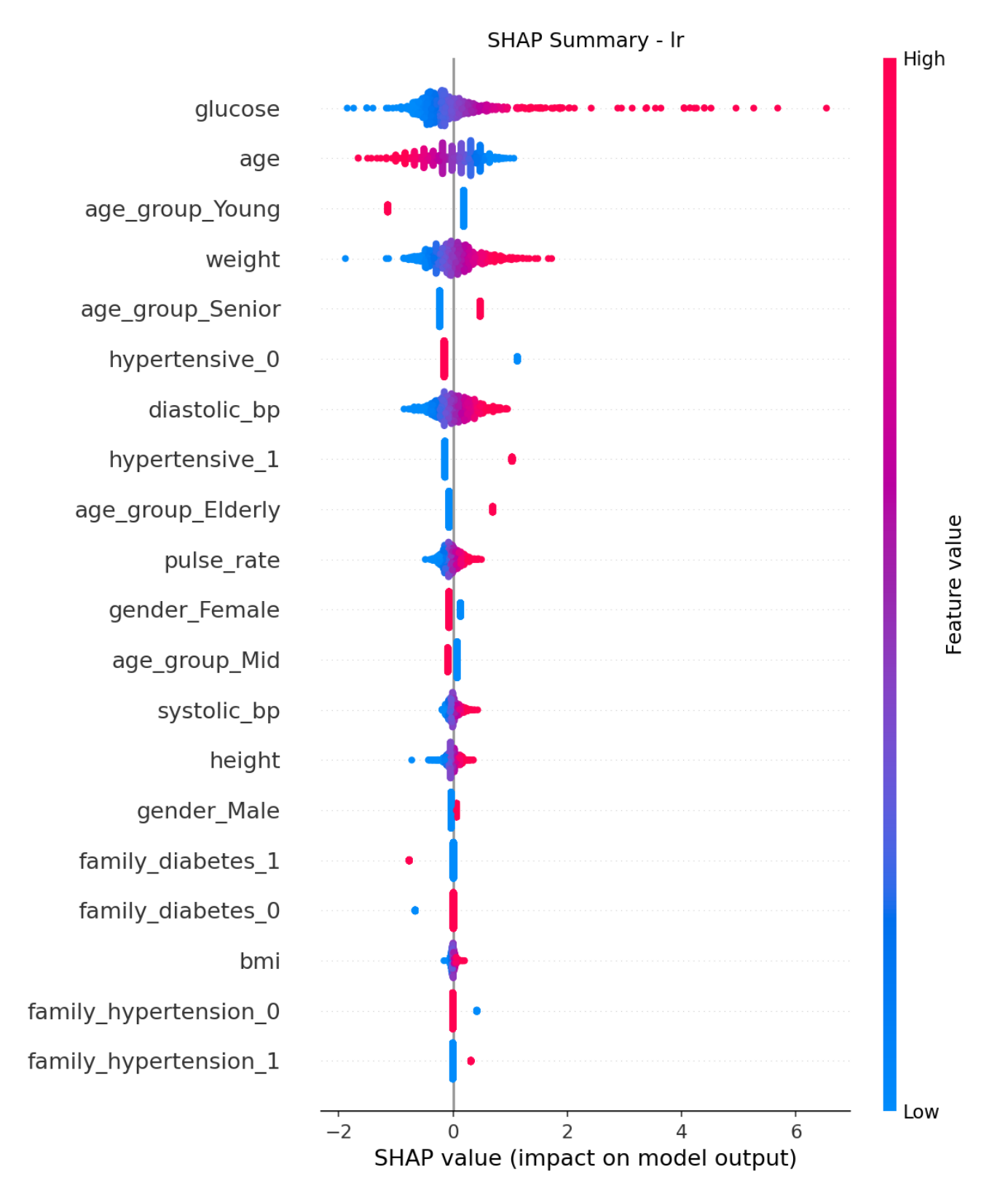
1. Global Explanations:

- SHAP summary plots
- Feature importance rankings

2. Local Explanations:

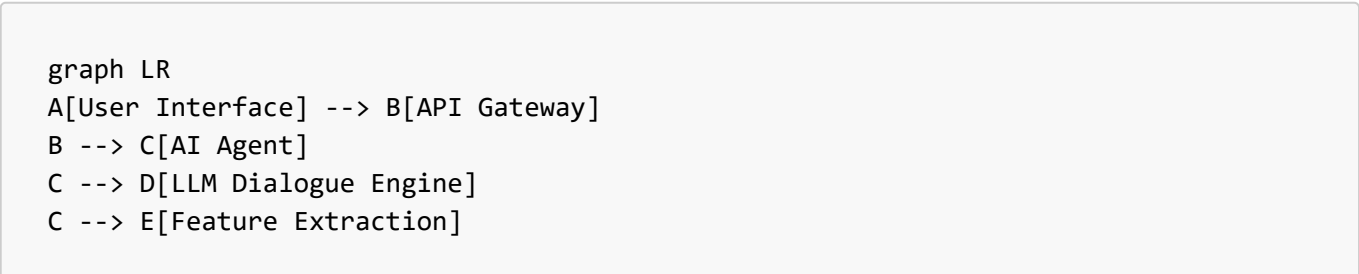
- LIME for individual predictions
- Interactive prediction breakdowns

```
# SHAP Implementation
explainer = shap.TreeExplainer(model.best_estimator_)
shap_values = explainer.shap_values(X_test)
shap.summary_plot(shap_values, X_test)
```

Glucose levels and age identified as top predictors

System Architecture



```
E --> F[Trained Models]
F --> G[Prediction Engine]
G --> H[Feedback Mechanism]
H --> I[Model Refinement]
```

Conclusion

This system demonstrates:

- 95.97% prediction accuracy using XGBoost
- Effective class imbalance handling via SMOTENC
- Seamless integration of conversational AI with predictive analytics
- Transparent decision-making through XAI techniques
- Potential for deployment in clinical support systems

 System Demo